

Deep Learning & Neural Network

- ANN (Artificial Neural Network)
- CNN (Convolutional Neural Network)

Machine Learning

Raw Data -> Preprocess, clean, Scale -> Feature Engineering -> Model -> output

Deep Learning

Raw Data -> Preprocess ,clean, Scale -> neural Network -> Otuput

Deep learning is the subfield of machine learning that uses artificial neural network (ANN) with many layers (hence "deep") to learn patterns directly from raw data(images, text, audio etc.).

- Deep learning removes the need for manual feature engineering not preprocessing.
- Feature engineering *(which DL eliminate) is the human deciding which pattern matter and computing them by hand- "Extract the edges" "count the syllables".

Types of Neural Network

1. FNN(Feed Forward Nueral Network)/MLP(Multi layer perceptron)

- It is good for structured data/ Tabular data and for simple regression/classification

2. Convolutional Neural Network(CNN):

- Build for grid like data , images, Instead connecting every pixel to every neuron(which would be enormous). A CNN slides a small filters accross image to detect local patterns - edges, textures, shape and object.

3. Recurrent Neural Network(RNN):

- Built for sequential data -text, speech, time-series , where order matters. Unlike a feedforward network , A RNN has a loop: It feeds its own previous output back as input, giving it a form of memory, of what come before.
- It's weakness is forgetting long-range context, which led to improve variants.

4. LSTM(Long short term memory)(RNN variant) and GRU(Gated Recurrent unit network)- RNN with gates that decide what to remember and what to forget letting them handle the layer sequence.

5. Transformers:

- The architecture behind the modern language models(GPT,BERT, and Claude). It replaced RNNs for most sequence tasks by using a mechanism called "attention", which lets the model look at all parts of sequence at once and weigh which parts matter most rather than processing word by word.

6. Auto Encoders:

- An unsupervised network that learns to compress data into a small representation and then reconstruct it. Used for dimensionality reduction, anomaly detection
- It has two halves: An encoder(compresses) and a Decoder (Rebuild).

7. Generative Adversarial Network(GAN):

- Two networks trained against each other: *Generator* that creates fake data and a *Discriminator* that tries to spot the fakes. The competition until the generator produces highly realistic output. Used to generating images, deepfakes and synthetic data.